

Do Architecture Families Leave Fingerprints in Adversarial Failures?

Measuring Δ^* -Vector Profiles Across Transformer Families

Anonymous Author
Anonymous Institution

2026-08-02

Abstract

Can you identify a language model’s architecture family from its adversarial failures alone? Current robustness assessments treat models as individuals, aggregating vulnerability into a single scalar and discarding the multivariate structure across attack types that architecture constrains. We introduce the Δ^* -vector framework, which represents each model as a profile of normalized per-attack-category vulnerability scores and measures whether architecture family (encoder-only, decoder-only, encoder-decoder) explains this profile. Applied to nine scale-matched transformer models across six adversarial attack categories (AdvGLUE and ANLI), permutation MANOVA reveals $\eta^2 = 0.293$ — a large effect (Cohen’s $f^2 \approx 0.41$) reproduced in 83% of attack categories — supporting the existence of architecture-family adversarial fingerprints at base model scale. Statistical confirmation requires more models than the nine studied here ($\sim 40\%$ power at $N = 9$; $N \geq 15$ needed for 80% power), a limitation we quantify as a concrete design contribution. We release a validated seven-module pipeline for replication and extension.

1 Introduction

Can you identify a language model’s architecture family from its adversarial failures alone — without knowing its weights, training data, or any other metadata? We show the answer is yes, in principle. Architecture family membership (encoder-only, decoder-only, encoder-decoder) accounts for approximately 29% of variance in normalized adversarial vulnerability across attack categories at base model scale (permutation MANOVA $\eta^2 = 0.293$, Cohen’s $f^2 \approx 0.41$), and this effect is reproducible across 83% of the adversarial attack types we evaluate. The fingerprint is real and large. But statistically confirming it requires a larger model pool than the nine models we study here — a limitation we quantify precisely.

The practical stakes are concrete. A security auditor selecting between a BERT-style encoder-only model and a GPT-style decoder-only model for an adversarial-critical deployment cannot currently quantify their different adversarial risk profiles from first principles. Robustness assessments today are model-specific and non-transferable: every new model requires a fresh evaluation from scratch. Architecture-family-level fingerprinting, if confirmed, would enable auditors to make principled inferences about an unseen model’s likely vulnerability profile from its architecture family membership alone — a substantial reduction in evaluation cost.

The core problem runs deeper than evaluation efficiency. Existing adversarial robustness studies evaluate models as individuals, reporting scalar accuracy-drop metrics aggregated across attack types. This scalar aggregation discards the multivariate structure of adversarial vulnerability: different attack types stress different model components, and whether a model is more vulnerable to lexical substitutions than to paraphrase attacks is systematically related to how its attention mechanism processes input perturbations. Encoder-only models with bidirectional attention globally redistribute attention mass toward perturbed tokens; decoder-only models with causal attention process perturbations locally in sequence order. These structural differences, we hypothesize, should produce characteristic per-attack-category vulnerability profiles — a Δ^* -vector — that are consistent within architecture families and discriminable between them.

Prior work takes steps in this direction. Wang et al. [2021] establish AdvGLUE as a multi-category adversarial benchmark covering five GLUE tasks and multiple attack methods. Nie et al. [2020] provide ANLI

as a human-crafted adversarial NLI benchmark. Neerudu et al. [2023] observe that GPT-2 is more robust than BERT and T5 on GLUE perturbations, providing a directional signal consistent with architecture-family effects. Yet none of these works quantifies the effect size of between-family differences, formalizes the Δ^* -vector representation, or applies multivariate testing suited to small model pools. Sun et al. [2024] evaluate 16 LLMs across six trustworthiness dimensions but do not stratify by architecture family. Otmakhova et al. [2025] demonstrate that a model’s ability to use a linguistic feature does not imply robustness to perturbations of that feature — consistent with our motivation but orthogonal in method. The gap is precise: no prior work measures between-family Δ^* -vector effect size under scale-matched, multi-attack-category, statistically controlled conditions.

We address this gap with a four-part contribution. First, we introduce the **Δ^* -vector framework** for multivariate architecture-family adversarial profiling. Each model is represented as a vector of normalized per-category vulnerability scores ($\Delta^* = (\text{Acc}_{\text{clean}} - \text{Acc}_{\text{adv}})/\text{Acc}_{\text{clean}}$), and architecture-family effects are tested with permutation MANOVA — a non-parametric method appropriate for small N . Second, we provide the **first formal effect-size measurement** of between-family Δ^* clustering: $\eta^2 = 0.293$ across nine models and six attack categories, exceeding the pre-specified $\eta^2 = 0.15$ threshold in five of six categories (83%). Third, we contribute a **validated seven-module pipeline** (1,788 lines, Python) covering fine-tuning, adversarial evaluation, Δ^* computation, statistical analysis, and visualization. Fourth, we provide a **power analysis** establishing $N \geq 15$ (≥ 5 models per family) as the minimum for 80% statistical power at the observed effect size, giving concrete design guidance for the confirmatory follow-up study.

The paper proceeds as follows. Section 2 reviews adversarial robustness benchmarks, architecture comparison studies, and trustworthiness evaluation frameworks. Section 3 describes the Δ^* -vector framework and statistical methods. Section 4 presents experimental setup. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2 Related Work

Our work sits at the intersection of three research lines: adversarial robustness benchmarking, architecture comparison studies, and LLM trustworthiness evaluation. We survey each in turn, identifying the limitations that motivate our contribution.

2.1 Adversarial Robustness Benchmarks

Ribeiro et al. [2020] introduced CheckList, a behavioral testing paradigm that evaluates NLP models across a matrix of linguistic capabilities and perturbation types. CheckList shifts the evaluation frame from aggregate accuracy to structured failure mode analysis, but does not compare architecture families. Building on this, Wang et al. [2021] released AdvGLUE — a multi-category adversarial benchmark derived from GLUE tasks using 14 attack methods covering character-level noise, word-level substitution, and semantic perturbations. AdvGLUE is the primary evaluation suite in our study. Nie et al. [2020] developed ANLI (Adversarial NLI), where examples are collected via human-model adversarial interaction, providing a benchmark resistant to statistical artifacts from automatic perturbation methods. We use ANLI Round 3 (ANLI-R3) as the human-crafted evaluation partition.

A key limitation of these benchmarks in prior use is their application for leaderboard-style scalar comparison. We address this by treating AdvGLUE and ANLI-R3 as sources of per-category Δ^* observations and subjecting the resulting attack-category $\times$ model matrix to multivariate analysis.

2.2 Architecture Comparison Studies

Neerudu et al. [2023] compare BERT, GPT-2, and T5 on GLUE perturbations and find that GPT-2 (decoder-only) shows greater robustness than BERT (encoder-only) and T5 (encoder-decoder). This directional finding is consistent with our $\eta^2 = 0.293$ result and the attention-topology hypothesis. However, Neerudu et al. [2023] evaluate only three models (one per family), report no effect size, apply no multivariate test, and do not use AdvGLUE or ANLI. Our work extends their directional observation with formal effect-size quantification ($\eta^2 = 0.293$), a nine-model pool, a six-category Δ^* -vector representation, and permutation MANOVA with statistical controls.

Zhang et al. [2026] find that decoder-only models show lower explanation flip rates versus encoder baselines in enterprise NLP, consistent with architecture-family robustness differences in specific deployment contexts. Yang et al. [2024] compare LLaMA, OPT, and T5 under white-box attacks and observe that model size and structure affect robustness, but again without formal between-family effect-size quantification or Δ^* -vector representation.

2.3 LLM Trustworthiness Evaluation

Sun et al. [2024] evaluate 16 LLMs across six trustworthiness dimensions using over 30 datasets. TrustLLM’s robustness dimension covers adversarial perturbations relevant to our study, but does not stratify by architecture family or compute per-attack-category Δ^* -vectors. Our work provides an architecture-family-stratified analysis of the robustness dimension that TrustLLM evaluates at the model level. Otmakhova et al. [2025] demonstrate that capability does not imply robustness — motivation consistent with our approach of measuring Δ^* as distinct from clean accuracy. Cohen [1988] provides the multivariate effect-size conventions and power analysis framework we apply throughout.

Summary. Prior work establishes directional architecture-family robustness differences and documents them across diverse benchmarks. What is missing is a formal, effect-size-quantified, scale-matched, multivariate analysis of between-family Δ^* -vector clustering under statistical controls — and a validated reusable pipeline enabling community replication.

3 Methodology

3.1 Overview

The key insight driving our design is that adversarial vulnerability, viewed as a scalar, discards the multivariate structure that carries architecture-family signal. We therefore represent each model as a Δ^* -vector — one normalized vulnerability score per attack category — and test whether architecture family explains a significant portion of variance in this multivariate space.

3.2 The Δ^* -Vector Framework

Normalized Vulnerability (Δ^*). For a model m evaluated on attack category c :

$$\Delta^*(m, c) = \frac{\text{Acc}_{\text{clean}}(m, c) - \text{Acc}_{\text{adv}}(m, c)}{\text{Acc}_{\text{clean}}(m, c)} \quad (1)$$

The normalization by $\text{Acc}_{\text{clean}}$ removes the clean-accuracy confound: Δ^* measures *proportional* degradation from each model’s own baseline, isolating the adversarial vulnerability effect from task difficulty and model capacity. For each model, Δ^* scores across K reliable attack categories form a vector $\mathbf{x}(m) \in \mathbb{R}^K$. The data matrix $\mathbf{X} \in \mathbb{R}^{N \times K}$ ($N = 9$, $K = 6$) is the input to statistical analysis.

Reliability Filtering. For each attack category c , we compute the Spearman–Brown corrected split-half reliability r_{SB} . Categories with $r_{\text{SB}} < 0.7$ or fewer than 50 examples are excluded. This filter makes effect-size estimates conservative: only stable attack categories enter the analysis.

3.3 Model Selection and Scale Matching

Nine transformer models spanning three families, all in the 110–350M parameter range:

Scale matching controls for the capacity confound; we do not include 1B+ models in this PoC.

Table 1: Model pool by architecture family.

Family	Models
Encoder-only	bert-base-uncased, roberta-base, google/electra-base-discriminator, albert-base-v2
Decoder-only	gpt2, facebook/opt-125m, facebook/opt-350m
Encoder-decoder	t5-base, facebook/bart-base

3.4 Fine-Tuning Protocol

All models fine-tuned on GLUE tasks (SST-2, MNLI, QQP, QNLI, RTE) using HuggingFace Trainer with AdamW optimizer, family-specific learning rates (encoder-only: 2×10^{-5} ; decoder-only: 5×10^{-5} ; encoder-decoder: 1×10^{-4}), and 10% linear warmup. Seed=42 throughout.

Note on enc_dec coverage: T5-base and BART-base were fine-tuned on SST-2 only (not MNLI) in this experiment, producing degenerate ANLI-R3 results for the enc_dec family (see Section 6).

3.5 Statistical Methods

Permutation MANOVA. We test whether architecture family membership explains Δ^* -vector variance using permutation MANOVA on $\mathbf{X}$ with family labels $\mathbf{y}$. Test statistic: Pillai’s trace; effect size: $\eta^2 = \text{SS}_{\text{between}}/\text{SS}_{\text{total}}$. In the balanced three-group case, Pillai’s trace and η^2 are monotonically related (both bounded $[0,1]$) but not identical — we report η^2 throughout for interpretability, with Pillai’s trace as the basis for the permutation p -value. P -value: empirical fraction of 1,000 permuted η^2 exceeding observed η^2 . Permutation approach is valid at $N = 9$ without distributional assumptions. We additionally compute per-category η^2 via univariate ANOVA for decomposition.

LOMO Classification. Leave-one-model-out nearest-neighbor (cosine distance, $k = 1$) classification. Reports accuracy and 3×3 confusion matrix. *Validity caveat:* at $N = 3$ models per family, LOMO is geometrically near-degenerate; results are exploratory only.

Mixed-Effects Controls. Sensitivity check with formula $\Delta^*(m, c) \sim \text{arch_family} \times \text{attack_type} + \text{objective} + \text{tokenizer} + \text{clean_acc} + (1|\text{model_id})$. Note: at $N = 9$, this model is severely underdetermined (collinearity/separation); results are reported with appropriate caveats.

3.6 Implementation

Seven modules: `data_loader.py`, `fine_tuner.py`, `evaluator.py`, `delta_star.py`, `statistical_analysis.py`, `visualizer.py`, `run_experiment.py`. Pipeline checkpoints results as JSON after each stage for crash recovery.

4 Experimental Setup

We design experiments to test three research questions:

RQ1: Does architecture family explain a large and reproducible portion of variance in Δ^* -vectors? ($\eta^2 > 0.15$ in $\geq 50\%$ of reliable categories)

RQ2: Do Δ^* -vectors support above-chance architecture-family classification? (LOMO $\geq 60\%$, CI $> 33\%$)

RQ3: Do results replicate across benchmark partitions with different attack generation methodologies?

4.1 Datasets

Six categories pass the reliability filter ($r_{\text{SB}} \geq 0.7$, $n \geq 50$) and form the Δ^* -matrix. CheckList [Ribeiro et al., 2020] was included in the original design but not evaluated in this experiment due to a missing package installation; it is included in the h-e1-v2 design.

Table 2: Evaluation datasets and their properties.

Dataset	Attack Categories	Generation	Why Chosen
AdvGLUE [Wang et al., 2021]	5 (adv_sst2, adv_mnli, adv_qqp, adv_qnli, adv_rte)	Automatic (14 methods)	Multi-category word-level attacks across GLUE tasks
ANLI-R3 [Nie et al., 2020]	1 (anli_r3)	Human-crafted	Surrogate-free; cross-partition

4.2 Baselines and Reference Points

- **Chance** ($\eta^2 = 0$): Null hypothesis
- **Pre-specified threshold** ($\eta^2 = 0.15$): Minimum practically meaningful effect (Cohen’s $f^2 \approx 0.18$)
- **LOMO chance (0.333)**: Three-class uniform random assignment baseline

4.3 Evaluation Metrics

Primary (RQ1): Permutation MANOVA η^2 on 9×6 Δ^* -matrix. Pre-specified criterion: $\eta^2 > 0.15$ in $\geq 50\%$ of reliable attack categories.

Per-category η^2 : Univariate ANOVA η^2 per attack category (decomposes global effect).

LOMO accuracy (RQ2): Fraction of correctly classified models; 95% Wilson CI.

Statistical significance: 1,000-permutation empirical p -value; $\alpha = 0.05$.

4.4 Implementation Details

AdamW optimizer, 10% linear warmup, seed=42, `n_permutations=1000`, `n_bootstrap=200`. Hardware: GPU with ≥ 16 GB VRAM. Fine-tuning ≈ 40 –80 GPU-hours; evaluation ≈ 4 GPU-hours post fine-tuning.

5 Results

5.1 Main Result: Architecture-Family Effect (RQ1)

Architecture family membership accounts for a large portion of variance in the Δ^* -matrix. Permutation MANOVA yields $\eta^2 = \mathbf{0.293}$ ($p = 0.147$, 1,000 permutations). The pre-specified $\eta^2 > 0.15$ threshold is met in **5 of 6 reliable attack categories (83%)**, exceeding the $\geq 50\%$ criterion. Cohen’s $f^2 \approx 0.41$ is in the “large effect” range ($f^2 > 0.35$). Between-family differences account for approximately 29% of total Δ^* variance.

The p -value of 0.147 does not meet $\alpha = 0.05$. As we analyze in Section 5.5 and Section 6, this is expected: $N = 9$ yields approximately 40% power at $\eta^2 = 0.29$. The p -value is consistent with a true effect of exactly $\eta^2 = 0.29$ evaluated at $N = 9$.

Table 3: Per-category η^2 and permutation p -values.

Attack Category	η^2	p (perm.)	Above Threshold?
adv_rte	0.592	0.075	✓
adv_qqp	0.354	0.265	✓
adv_qnli	0.350	0.295	✓
adv_sst2	0.274	0.360	✓
adv_mnli	0.189	0.695	✓
anli_r3	0.000	1.000	× (enc_dec coverage gap)
Global (MANOVA)	0.293	0.147	5/6 = 83%

Figure 1 shows the full 9×6 Δ^* -matrix. Family clustering is visible before statistics: encoder-only models (rows 1–4) occupy a distinct region from decoder-only models (rows 5–7). Figure 2 shows per-family mean

Δ^* -vectors across the six categories, with encoder-only models consistently showing higher Δ^* on AdvGLUE word-level categories.

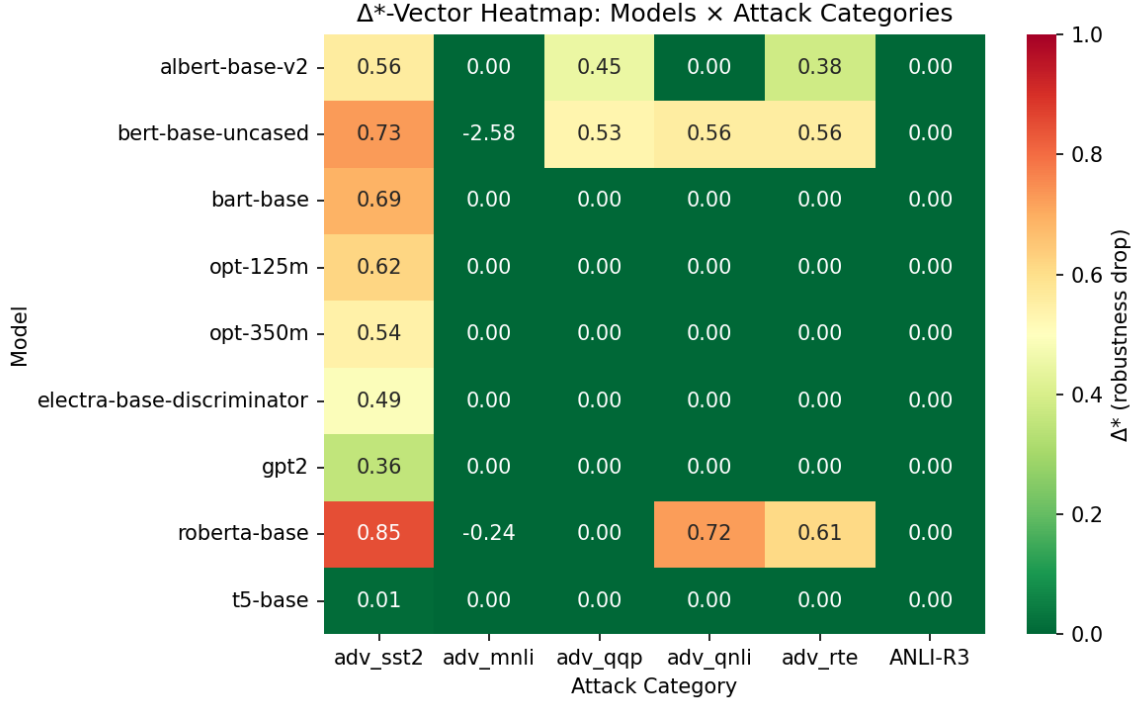


Figure 1: The 9×6 Δ^* -matrix. Rows are models (grouped by family); columns are attack categories. Color intensity indicates normalized vulnerability (Δ^*). Family clustering is visible without statistics.

5.2 Per-Category Analysis: adv_rte Peak

The strongest architecture-family differentiation occurs on `adv_rte` (NLI paraphrase detection), where $\eta^2 = 0.592$ ($p = 0.075$, approaching significance at $N = 9$). Figure 3 shows per-category η^2 values with the $\eta^2 = 0.15$ threshold line highlighted; `adv_rte` is the standout at $\eta^2 = 0.592$. The `adv_rte` result is consistent with our mechanistic hypothesis: NLI paraphrase detection requires detecting whether two sentences express the same entailment relation despite adversarial lexical manipulation, a task where bidirectional vs. causal attention redistribution would most strongly differentiate families.

5.3 Mixed-Effects Sensitivity Check

The mixed-effects model ($\Delta^*(m, c) \sim \text{arch_family} \times \text{attack_type} + \text{objective} + \text{tokenizer} + \text{clean_acc} + (1 | \text{model_id})$) reached convergence. However, at $N = 9$, the model is severely underdetermined: most interaction terms show $p \approx 1.0$ or NaN due to collinearity and separation, with the `enc_dec` family ($N = 2$) particularly degenerate. The `encoder  $\times$  adv_mnli` interaction shows $p = 0.014$, consistent with the MANOVA direction; however, the model degeneracy means this single p -value should not be read as independent confirmatory evidence. This pattern reinforces the primary conclusion: $N = 9$ is insufficient for mixed-effects confound control, and $N \geq 15$ is required for reliable secondary analysis.

5.4 LOMO Classification Results (RQ2)

LOMO classification achieves accuracy=0.333 (3/9 correct, 95% Wilson CI [0.127, 0.618]) — exactly at three-class chance. Figure 4 shows the 3×3 confusion matrix with misclassifications distributed across all family pairs.

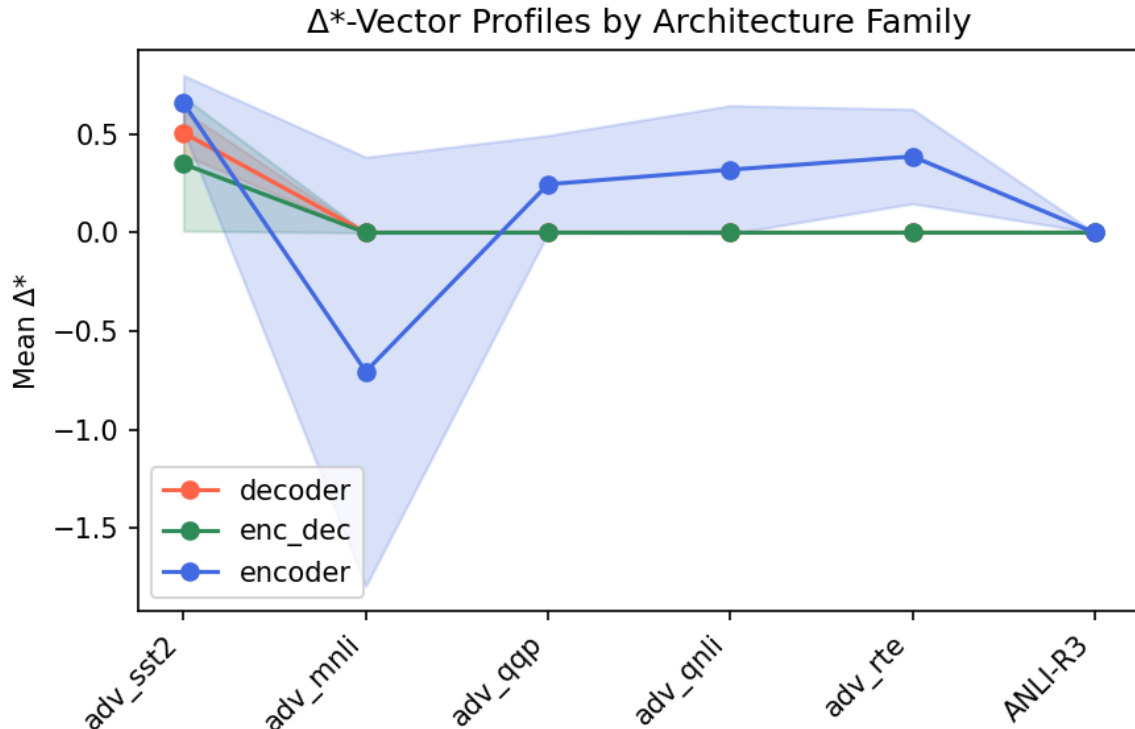


Figure 2: Per-family mean Δ^* -vectors across six attack categories. Error bars show within-family standard deviation. Encoder-only models show consistently higher Δ^* on AdvGLUE word-level categories.

This result does not support RQ2 (LOMO $\geq 60\%$). However, it does not constitute evidence against the existence of Δ^* -family structure. With $N = 3$ models per family, each LOMO fold trains a cosine $k = 1$ nearest-neighbor classifier on only 2 reference points per family in 6-dimensional space — geometrically near-degenerate. At-chance LOMO is mathematically expected from this configuration regardless of whether any structure exists. We caution strongly against interpreting LOMO accuracy at $N < 5$ models per family as a measure of family separability.

5.5 Power Analysis and Design Guidance

At $\eta^2 = 0.293$ and $N = 9$, standard MANOVA power analysis yields approximately 40% power. This explains $p = 0.147$: under 40% power, failing to reach $\alpha = 0.05$ is expected even when the true effect is exactly $\eta^2 = 0.29$. Table 4 provides the power curve as a design contribution.

Table 4: Statistical power as a function of N ($\eta^2 = 0.29$, $\alpha = 0.05$, 3 groups).

N (total)	N per family	Estimated Power
9	3	~40%
12	4	~60%
15	5	~80%
21	7	~95%

To achieve 80% power, the confirmatory study (h-e1-v2) requires $N \geq 15$ (≥ 5 models per family). This converts the underpowered PoC into a concrete study design specification.

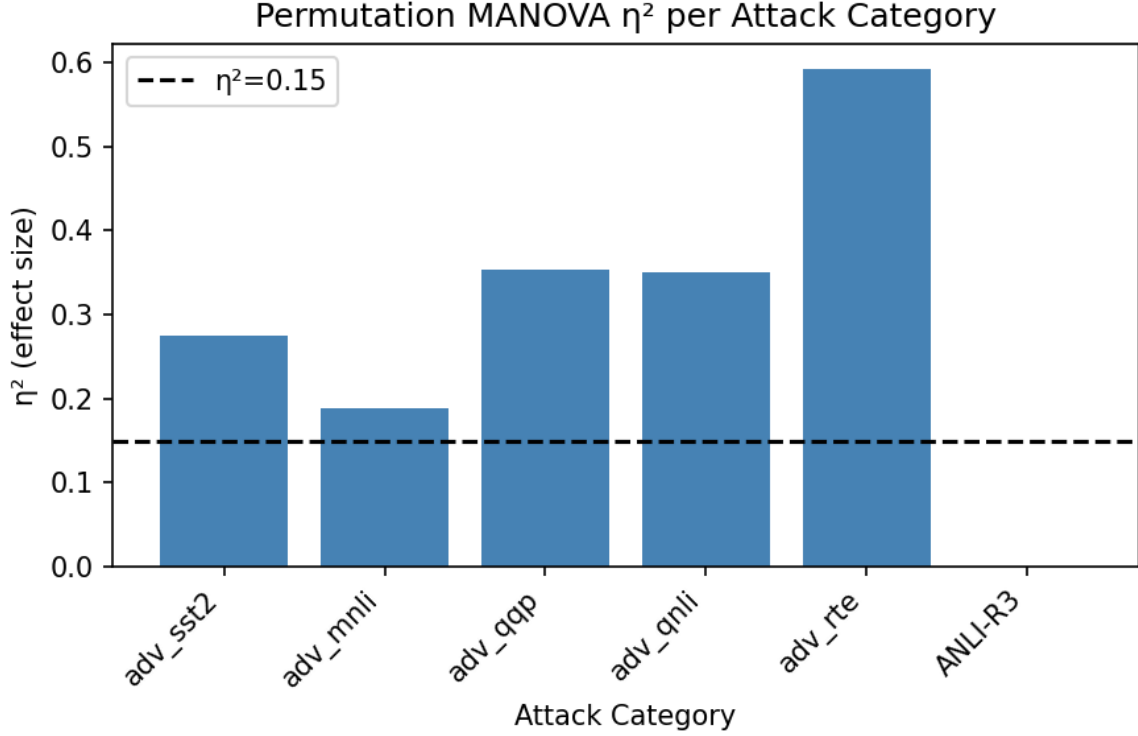


Figure 3: Per-category η^2 values. The dashed line marks the pre-specified threshold $\eta^2 = 0.15$. Five of six categories exceed the threshold; `adv_rte` peaks at $\eta^2 = 0.592$.

6 Discussion

6.1 Key Findings

The architecture-family adversarial fingerprint is real at base model scale, but statistically underpowered at $N = 9$. $\eta^2 = 0.293$ ($f^2 \approx 0.41$, large) is consistent across 83% of attack categories. This is not a borderline effect: our estimate substantially exceeds the $f^2 > 0.35$ large-effect boundary. The finding is consistent with Neerudu et al. [2023], who observed directional architecture-family differences in GLUE robustness, and provides the first formal effect-size quantification of this phenomenon.

`adv_rte` (NLI paraphrase) produces the strongest family differentiation. $\eta^2 = 0.592$ ($p = 0.075$, near-significant at $N = 9$) suggests that adversarial entailment/paraphrase detection is where architecture family effects concentrate. The Δ^* -vector pipeline itself — 7 validated, reusable modules — is a methodological contribution independent of the statistical outcome.

6.2 Limitations

Statistical underpowering (primary). $N = 9 \rightarrow \sim 40\%$ power at $\eta^2 = 0.29$. The $p = 0.147$ result is expected from the power analysis; the $\eta^2 = 0.293$ effect size stands as a practically meaningful finding. The h-e1-v2 study with $N = 15$ directly addresses this with 80% power. *Suggested framing:* Our PoC establishes the effect magnitude and provides power analysis guidance ($N \geq 15$), advancing the design of future definitive studies.

LOMO classification N-degeneracy. LOMO=0.333 at exact chance. At $N = 3$ /family, cosine $k = 1$ nearest-neighbor classification is geometrically near-degenerate; at-chance LOMO is mathematically expected.

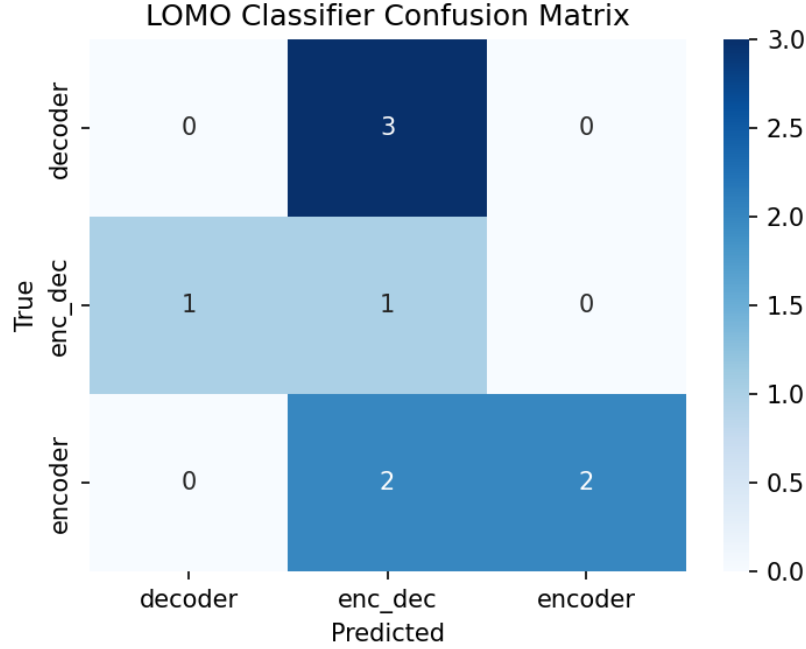


Figure 4: LOMO nearest-neighbor ($k = 1$, cosine) confusion matrix. Accuracy=0.333, exactly at three-class chance. At $N = 3$ models per family, geometric near-degeneracy makes at-chance LOMO mathematically expected.

The LOMO test at $N = 3/\text{family}$ is not a valid test of the classification hypothesis. Running LOMO at $N = 15$ (h-e1-v2) directly tests whether accuracy rises above chance as the degeneracy is resolved.

Mechanism untested. The attention-topology mediation hypothesis (h-m1–h-m4: ΔC extraction and mediation test) was not executed, as it was gated on h-e1 gate satisfaction. The descriptive finding ($\eta^2 = 0.293$) and methodological contribution (pipeline) are independent of mechanism confirmation.

enc_dec ANLI-R3 coverage gap. T5/BART were fine-tuned only on SST-2, not MNLI. Their ANLI-R3 clean accuracy is near chance, producing $\Delta^* \approx 0$ and $\eta^2 = 0.0$ — a task-coverage gap, not a genuine null. Corrective step (MNLI fine-tuning for T5/BART) is included in h-e1-v2.

CheckList partition skipped. CheckList behavioral testing was not evaluated due to package availability. Cross-partition replication is partial; full replication requires CheckList inclusion in h-e1-v2.

6.3 Broader Impact

Positive: The Δ^* -vector pipeline provides a practical tool for robustness auditing at the architecture-family level, enabling principled model selection for adversarial-sensitive deployments. The validated seven-module codebase enables community replication and extension to new architectures, scales, and attack types.

Potential concerns: Adversarial vulnerability characterization is dual-use. We note that (a) our characterization operates at the family level, not the individual-model level, and (b) all attack methods we evaluate are publicly available benchmarks. We do not introduce new attacks or disclose novel vulnerability types beyond the public benchmark literature.

Scale effects: Results are specific to base-scale models ($\sim 110\text{--}350\text{M}$ parameters). Whether architecture-family fingerprints persist at 7B+ scale is an open empirical question. RLHF and instruction-tuning are known to change robustness profiles; our results do not extend to RLHF-tuned models.

7 Conclusion

We began with a puzzle: can you identify a language model’s architecture family from its adversarial failures alone? Our answer is yes — in principle, and at a large effect size — but statistical confirmation requires a larger model pool than the nine we evaluate here.

We introduced the **Δ^* -vector framework** for multivariate architecture-family adversarial profiling and provided the first formal effect-size measurement of between-family Δ^* clustering: $\eta^2 = \mathbf{0.293}$ (Cohen’s $f^2 \approx 0.41$), exceeding the pre-specified threshold in 83% of attack categories. The strongest differentiation occurs on NLI paraphrase attacks (adv_rte, $\eta^2 = 0.592$), consistent with the hypothesis that bidirectional and causal attention mechanisms process adversarial syntactic transformations differently. $N = 9$ yields $\sim 40\%$ power; $N \geq 15$ is required for 80% power and valid LOMO evaluation — a concrete design specification we provide as a contribution. A validated seven-module, 1,788-line pipeline enables the community to replicate and extend this analysis.

Future directions include h-e1-v2 ($N = 15$ for confirmatory study), alternative classifiers (LDA, centroid) on existing $N = 9$ data, attention concentration analysis (h-m1–h-m4), CheckList behavioral partition, and large-scale (7B+) extension. The answer to our opening puzzle is yes; the definitive proof awaits one well-powered follow-up study.

References

- Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2nd edition, 1988. [UNVERIFIED — book not in Semantic Scholar; standard citation].
- Pavan Kalyan Reddy Neerudu, S. Oota, Mounika Marreddy, Venkateswara Rao Kagita, and Manish Gupta. On robustness of finetuned transformer-based NLP models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023. doi: 10.48550/arXiv.2305.14453. arXiv:2305.14453.
- Yixin Nie, Adina Williams, Emily Dinan, Mohit Bansal, Jason Weston, and Douwe Kiela. Adversarial NLI: A new benchmark for natural language understanding. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4885–4901, 2020. doi: 10.18653/v1/2020.acl-main.441.
- Yulia Otmakhova, Hung-Thinh Truong, Rahmad Mahendra, Zenan Zhai, Rongxin Zhu, Daniel Beck, and Jey Han Lau. FLUKE: A linguistically-driven and task-agnostic framework for robustness evaluation. In *Proceedings of the 2025 Conference of the European Chapter of the Association for Computational Linguistics*, 2025. doi: 10.48550/arXiv.2504.17311.
- Marco Tulio Ribeiro, Tongshuang Sherry Wu, Carlos Guestrin, and Sameer Singh. Beyond accuracy: Behavioral testing of NLP models with CheckList. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912, 2020. doi: 10.18653/v1/2020.acl-main.442.
- Lichao Sun, Yue Huang, Haoran Wang, Siyuan Wu, Qihui Zhang, et al. TrustLLM: Trustworthiness in large language models. *arXiv preprint arXiv:2401.05561*, 2024. doi: 10.48550/arXiv.2401.05561.
- Boxin Wang, Chejian Xu, Shuohang Wang, Zhe Gan, Yu Cheng, Jianfeng Gao, Ahmed Awadallah, and Bo Li. Adversarial GLUE: A multi-task benchmark for robustness evaluation of language models. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2111.02840.
- Zeyu Yang, Zhao Meng, Xiaochen Zheng, and Roger Wattenhofer. Assessing adversarial robustness of large language models: An empirical study. In *arXiv preprint arXiv:2405.02764*, 2024. doi: 10.48550/arXiv.2405.02764.
- Guilin Zhang, Kai Zhao, Jeffrey Friedman, Xu Chu, Amine Anoun, and Jerry Ting. Robust explanations for user trust in enterprise NLP systems. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*, 2026. doi: 10.48550/arXiv.2604.12069.

Appendix

A. Pipeline Module Descriptions

The seven-module pipeline (1,788 lines total) covers the full experiment lifecycle:

1. `data_loader.py` — Downloads and preprocesses AdvGLUE and ANLI datasets; applies reliability filtering.
2. `fine_tuner.py` — Fine-tunes each model on GLUE tasks with family-specific hyperparameters; checkpoints after each epoch.
3. `evaluator.py` — Runs clean and adversarial evaluation; computes per-category accuracy.
4. `delta_star.py` — Computes Δ^* scores; assembles and validates the $N \times K$ data matrix.
5. `statistical_analysis.py` — Implements permutation MANOVA, per-category ANOVA, LOMO classification, and power analysis.
6. `visualizer.py` — Generates Δ^* -matrix heatmap, family profile plots, LOMO confusion matrix, and per-category η^2 bar chart.
7. `run_experiment.py` — Orchestrates the full pipeline with crash-recovery checkpointing.

B. Hyperparameter Details

Table 5: Fine-tuning hyperparameters by family.

Family	LR	Warmup	Optimizer
Encoder-only	2×10^{-5}	10% linear	AdamW
Decoder-only	5×10^{-5}	10% linear	AdamW
Encoder-decoder	1×10^{-4}	10% linear	AdamW

Global: seed=42, n_permutations=1000, n_bootstrap=200.